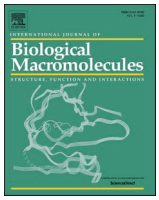
International Journal of Biological Macromolecules 330 (2025) 147895



Contents lists available at ScienceDirect

International Journal of Biological Macromolecules

journal homepage: www.elsevier.com/locate/ijbiomac



Fabrication of ternary deep eutectic solvent choline chloride:*p*-toluene sulfonic acid:Glycerol for pretreating wheat straw to augment cellulose accessibility, xylan removal, and lignin removal

Zhuo Zhang [1], Qianqian Gao [1], Yu-Cai He [*]

School of Pharmacy & School of Biological and Food Engineering, Changzhou University, Changzhou, 213164, China

ARTICLE INFO

Keywords:
Cellulose accessibility
Lignin removal
Xylan removal

ABSTRACT

In this research, a novel ternary deep eutectic solvent (TDES) ChCl:*p*-TsOH:Gly (1:1:1), composed of choline chloride (ChCl), *p*-toluene sulfonic acid (*p*-TsOH), and glycerol (Gly), was successfully fabricated and adopted to pretreat wheat straw under relatively mild conditions (80 °C, 45 min) for removing xylan (99.9 %), eliminating lignin (88.5 %) and elevating subsequent enzymatic saccharification efficiency (98.1 %). The combined hydrolysis factor and combined delignification factor quantitatively predicted the dissolution kinetics of residual xylan and residual lignin, respectively, with high accuracy. Following ChCl:*p*-TsOH:Gly treatment, the accessibility of cellulose in wheat straw elevated considerably to 691.5 mg dye/g cellulose, while surface lignin area diminished to 134.6 m^2/g and lignin hydrophobicity was weakened to 2.6 L/g. These structural and surface property modifications contributed to the enhancement of enzymolysis. Molecular dynamics simulations and quantum chemical calculations were adopted to reveal van der Waals interactions between ChCl:*p*-TsOH:Gly and lignin, as well as dispersion interactions between ChCl and lignin, consequently elucidating the molecular-level interaction mechanism between lignocellulose and ChCl:*p*-TsOH:Gly. In summary, this study achieved effectual pretreatment of wheat straw using the TDES ChCl:*p*-TsOH:Gly but also considerably promoted the depolymerization of lignocellulosic biomass in biorefining processes, offering new insights into the value-added utilization of biomass resources.

1. Introduction

To date, the development of clean energy alternatives has emerged as a critical priority [1,2]. Lignocellulosic biomass exhibits considerable potential for manufacturing bio-based materials, bioethanol, and biodiesel [3,4]. Wheat straw (WS), a widely available byproduct of global wheat cultivation, is often treated as waste, yet it contains a significant amount of polysaccharides, primarily composed of approximately 70 % sugar compounds, including cellulose, hemicellulose, and lignin [5,6]. Nevertheless, the complex structural interactions between cellulose and hemicellulose hinder enzymatic digestion [7,8]. Accordingly, it is essential to develop and select appropriate pretreatment strategies to overcome the recalcitrance of lignocellulosic materials and facilitate their effective disintegration [9].

Recently, deep eutectic solvents (DESs) have emerged as promising green solvents to pretreat biomass due to their tailorable hydrogen-bonding networks, low toxicity, and excellent recyclability [10–12]. For instance, Li et al. reported an acidic binary DES composed of ChCl/oxalic acid (molar ratio 3:1) that achieved 84 % delignification and 96 % enzymolysis efficiency after optimized treatment (130 °C, 1 h) of rapeseed straw [13]. Similarly, Tang et al. demonstrated that pretreatment with an acidic binary DES system of cetyltrimethylammonium bromide/glycerol (molar ratio 1:6, 200 °C, 90 min) resulted in 46 % delignification and 53 % xylan removal from rapeseed straw [14]. Recent studies indicated that incorporating a third component (such as alcoholic compounds, Lewis acids) into binary DESs to form ternary DESs (TDESs) can considerably improve biomass pretreatment efficiency [15–17]. Introducing a third component markedly modulates the physicochemical features of the solvent system (such as viscosity and dissolving capacity), consequently improving mass transfer rates and overall pretreatment efficiency [18,19]. Functional groups like hydroxyls introduced by the third component can strengthen the hydrogen-

* Corresponding author.
E-mail addresses: heyucai2001@163.com, yucaihe@cczu.edu.cn (Y.-C. He).
[1] They contributed equally to this work.

https://doi.org/10.1016/j.ijbiomac.2025.147895
Received 10 August 2025; Received in revised form 11 September 2025; Accepted 23 September 2025
Available online 24 September 2025
0141-8130/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

bonding network, which not only facilitates lower melting points enabling milder pretreatment temperatures but also augments the effectual separation of biomass components [20–22]. Compared to conventional thermochemical methods which typically require reaction temperatures exceeding 200 °C to achieve effective biomass decomposition, TDES based approaches enable efficient biomass conversion under significantly milder conditions (e.g., 80–120 °C). This substantial reduction in operating temperature not only directly diminishes energy input requirements but also mitigates additional costs associated with high-temperature-resistant equipment, thereby achieving considerable energy savings. Furthermore, the mild reaction conditions minimize excessive degradation of high-value components within the biomass, enhancing selectivity toward target products. Collectively, these advantages position TDES-mediated processes as a more sustainable technological pathway for biomass valorization. For example, Zhao et al. fabricated a TDES using octadecyl dimethyl ammonium chloride, propionic acid, and *p*-toluenesulfonic acid (*p*-TsOH), achieving effectual pretreatment of sorghum straw under relatively mild conditions with an enzymolysis efficiency of 69.3 % [23]. Zhang et al. prepared a TDES from choline chloride (ChCl), N-(2-hydroxyethyl)ethylenediamine (NE), and iron(III) chloride for corn stover treatment under mild conditions, achieving 87 % delignification [24]. Similarly, He et al. reported that a TDES composed of ChCl, glycerol (Gly), and aluminum chloride removed 70.8 % lignin and 87.7 % hemicellulose from reed straw [25]. Consequently, TDES technology enables the effectual conversion of previously underutilized biomass resources, apparently broadening the scope of biorefinery feedstocks and opening new pathways toward enhancing the economic viability and sustainability of full biomass component utilization.

This work aimed to develop an effectual pretreatment process for wheat straw (WS) based on a DES system composed of choline chloride (ChCl), *p*-toluenesulfonic acid (*p*-TsOH), and glycerol (Gly), with the objective of enhancing its physicochemical properties and microstructural characteristics. The two-phase combined hydrolysis factor (CHF) model and the combined delignification factor (CDF) model can be utilized to quantitatively characterize the processes of hemicellulose dissolution and lignin removal, respectively. The effectiveness of the pretreatment can be estimated by analyzing variations in component content and enzymolysis performance, while key operational parameters—such as performance temperature, operation duration, and molar ratio—are systematically optimized. The response surface methodology can be adopted to determine the optimal DES-treatment conditions and to estimate the enzymatic digestion rate as a response variable. Systematically investigating the regulatory of the pretreatment on influencing cellulose accessibility, surface lignin coverage, and hydrophobicity can be implemented to assess the impacts on the subsequent enzymolysis efficiency. Fourier-transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), and other analytical techniques may be utilized to characterize the structural and compositional alterations in WS before and after DES treatment, consequently elucidating the mechanisms underlying lignin removal and enhanced enzymolysis. To explore the interaction mechanisms between biomass and DES at the molecular level, molecular dynamics simulations can be adopted, offering theoretical insights for the development of efficient biomass pretreatment strategies. Overall, this study presents a comprehensive analysis and in-depth discussion of the ChCl:*p*-TsOH:Gly-based biomass pretreatment process.

2. Materials and methods

2.1. Preparation of DESs

ChCl, *p*-TsOH, and glycerol were blended at varying molar ratios (1:1:0.5, 1:1:1, 1:1:2, 1:1:3). Concurrently, ChCl and *p*-TsOH were individually combined with ethylene glycol (EG) at a molar ratio of 1:1:1. The DES preparation was implemented at 80 °C for 15 to 30 min, and the obtained homogeneous and transparent liquids were used to pretreat WS without further purification.

2.2. Treatment of WS with DESs

In a three-necked flask, 60 g of TDES and 6 g of WS were mixed, resulting in a solid-to-liquid ratio of 1:10 (*w*/w). The mixture was stirred uniformly in an oil bath at a pre-determined temperature range of 40–120 °C for a variable duration of 30–75 min.

3. Results and discussion

3.1. The determination of K-T parameters of TDESs

The Kamlet-Taft parameters are empirical descriptors adopted to characterize the microscopic properties of solvent molecules and to quantitatively assess the specific interactions between solvents and solutes [26]. In this study, we systematically investigated the effects of EG) and Gly as hydrogen bond donors (HBDs) on the solvation properties of DES-ChCl. Specifically, EG and Gly were introduced into the DES-ChCl system at identical molar ratios, and the Kamlet-Taft parameters of the resulting binary DESs (designated as $TDES_{EG}$ and $TDES_{Gly}$, respectively) were accurately measured (Table 1). The experimental results clearly demonstrated that, compared with $TDES_{EG}$, $TDES_{Gly}$ exhibited significantly higher hydrogen bond donor ability (α value), hydrogen bond acceptor ability (β value), and dipolarity/polarizability (π* value). This notable difference in solvation properties is primarily attributed to the fundamental structural differences between Gly and EG—specifically, Gly contains an additional hydroxyl (-OH) functional group compared to EG. This additional hydroxyl group exerts a dual influence: first, it enhances the acidity of glycerol, providing more hydrogen bond donor sites and thereby directly increasing its hydrogen bond donor ability (α value); second, due to the high electronegativity of the oxygen atom, this hydroxyl group also significantly enhances Gly's capacity as a hydrogen bond acceptor (β value) [27]. Moreover, the molecular structure of glycerol is more complex, and its electron cloud distribution is more responsive to external electric fields, leading to greater polarizability, which is reflected in the significantly elevated π* value [27]. Given the overall advantages of $TDES_{Gly}$ in terms of α, β, and π* parameters—factors directly associated with the dissolution capacity, catalytic activity, and interaction strength with substrates (e.g., biomass) of DESs—Gly was selected as the optimal third component for constructing a novel TDES system in combination with ChCl and *p*-TsOH, based on its superior solvation characteristics.

The molar ratio of DES is a critical factor influencing the Kamlet-Taft parameters. In this work, $TDES_{Gly}$ was prepared with varying molar ratios (1:1:0.5, 1:1:1, 1:1:2, 1:1:3), and the impacts of these ratios on the Kamlet-Taft parameter values were systematically estimated. As the molar ratio of glycerol (Gly) elevated, the α and π* values of the Kamlet-Taft parameters gradually increased, whereas the β value showed a corresponding decrease (Table 1). The hydroxyl groups in glycerol are capable of forming hydrogen-bonds with the sulfonic acid oxygen atoms of *p*-TsOH or the cationic moiety of ChCl in DES-ChCl, consequently

Table 1
Kamlet-Taft parameters and viscosity of TDESs.

HBA	HBD	The third component	molar ratio	Kamlet-Taft parameters			Viscosity, mPa·s
				α	β	π*	
ChCl	PTSA	EG	1:1:1	1.85	1.35	0.11	349.63
ChCl	PTSA	Gly	1:1:0.5	1.87	1.35	0.13	349.24
			1:1:1	1.99	1.33	0.15	345.52
			1:1:2	2.63	1.30	0.16	526.70
			1:1:3	2.66	1.30	0.17	550.80

"Viscosity measurements of all DES samples were conducted at 25 °C."

improving the hydrogen-bond acidity (α value) of the DES system [28]. However, an excessive amount of glycerol introduces a higher number of hydroxyl groups that may compete with the intrinsic hydrogen-bond acceptor sites in DES-ChCl, realizing a diminishment in the system's overall proton-accepting and hydrogen-bond-forming ability and consequently lowering the β value [28]. Additionally, the occurrence of hydroxyl groups in glycerol can increase molecular deformability, making the system more susceptible to polarization under an external electric field. The results in an elevated π* value and a corresponding decrease in the cellulose dissolution capacity [27].

3.2. Investigating biomass chemical component variations under different ternary DES treatment

The primary objective of pretreatment is to overcome the structural and compositional barriers inherent to biomass, consequently augmenting the accessibility of biomass to reaction reagents by elevating the contact area and availability of reactive sites [29]. In this research, the impacts of two different TDES, namely $TDES_{EG}$ and $TDES_{Gly}$, on WS before and after DES treatment were estimated. Additionally, key pretreatment parameters—such as the molar ratio of DES components, temperature, and treatment duration—were optimized. As shown in Fig. 1, the glucan content in raw WS was 38.2 %. After treatment with $TDES_{EG}$ and $TDES_{Gly}$, the glucan content in the WS solid residue increased to 70.2 % and 73.3 %, respectively. These results indicate that the incorporation of a third component considerably enhanced the glucan content in WS following DES treatment. Compared to binary DES-ChCl alone, supplementing the third component to form TDES could improve the removal ability of WS xylan, resulting in a reduction in xylan content in the WS solid residue. This improvement may be attributed to the introduction of glycerol, which potentially alters the hydrogen-bond network structure of DES, consequently enhancing its capability to dissolve lignin and hemicellulose in WS [29]. As a result, the WS lignin and xylan were more effectually dissolved and removed, exposing the glucan that was originally embedded within the WS lignocellulosic matrix, leading to an increased glucan content [30].

$TDES_{Gly}$ had a remarkable impact on the retention of WS glucan and the removal of lignin and xylan (Fig. 1). Under identical treatment conditions (80 °C, 60 min), compared to treatment with DES (ChCl:*p*-TsOH, mol:mol 1:1), $TDES_{Gly}$ (ChCl:*p*-TsOH:Gly, mol:mol:mol 1:1:1) formed by supplementing Gly to this DES increased the glucan content in the DES-pretreated WS from 62.6 % to 73.3 %. The glucan recovery elevated from 35.8 % to 68.0 %. The removal rates of lignin and xylan

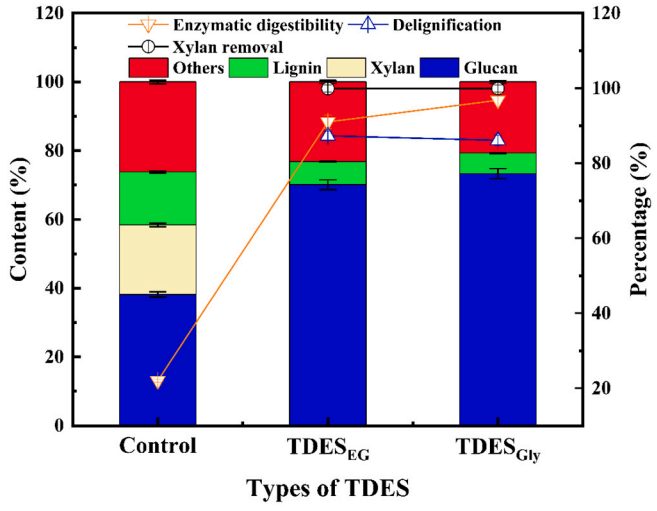


Fig. 1. Changes in the composition, enzymatic digestibility, xylan removal, and delignification of WS treated with different TDESs [molar ratio = 1:1:1, 80 °C, 60 min, and solid/liquid ratio = 1:10 *w*/w]. Control: Untreated WS.

were improved from 72.4 % and 90.5 % to 86.1 % and 99.9 %, respectively. The results manifested that supplementing glycerol not only effectually separated xylan and lignin from WS but also greatly retained glucan in WS. The hydroxyl groups in glycerol can form new hydrogen-bonds with other components in DES, which might promote the dissolution performance of DES for lignin and xylan, making them more easily dissolved from WS and consequently increasing lignin and xylan removal [31]. Accordingly, glycerol was selected as the third component to synthesize the ternary DES with DES-ChCl for the subsequent optimization of WS-pretreatment conditions.

The glycerol molar ratio in $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) may have an essential for impacting the pretreatment efficiency. $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) was adopted to pretreat WS (80 °C, 60 min) with four molar ratios (1:1:0.5, 1:1:1, 1:1:2, and 1:1:3). As depicted in Fig. 2a, upon increasing glycerol molar ratio in ChCl:*p*-TsOH:glycerol from 1:1:0.5 to 1:1:1, the glucan recovery elevated from 56.0 % to 68.0 %, and the separation ratio of lignin and xylan increased to 86.1 % and 99.9 %, respectively (Fig. 2a). However, increasing the molar ratio of glycerol in this ternary DES ChCl:*p*-TsOH:glycerol, the glucan content in WS and the separation ratio of lignin and xylan all diminished. As the molar ratio of glycerol was excessively high, the polarity of the system surpassed the optimal dissolution range for lignin and xylan, consequently diminishing their removal efficiency (Fig. 2b) [14]. The α value represents the capability of a solvent to donate protons and form hydrogen bonds with hydrogen-bond acceptors. As shown in Table 1, when the molar ratio of ChCl/*p*-TsOH/Gly elevated from 1:1:0.5 to 1:1:1, the α value increased significantly, indicating enhanced proton-donating capacity of the solvent, which consequently improved the pretreatment effectiveness. This was corroborated by the enzymatic hydrolysis efficiency, which increased from 92.5 % to 96.8 %, as depicted in Fig. 2b. Although further increasing the molar ratio could continue to elevate the α value, Table 1 displayed that the viscosity of ChCl/*p*-TsOH/Gly reached its minimum at a 1:1:1 molar ratio. Viscosity is a critical physical parameter influencing pretreatment performance. Excessively high viscosity impedes efficiency in multiple ways: it hinders the penetration and diffusion of the solvent into the internal pores of the biomass, reduces effective contact with components such as lignin and hemicellulose, and decelerates the reaction rate. Consequently, as illustrated in Fig. 2b, a continued increase in the molar ratio of ChCl/*p*-TsOH/Gly leads to a decline in enzymatic hydrolysis efficiency. An excessive amount of glycerol may also catalyze side-reactions such as cellulose dehydration and degradation, converting cellulose into non-glucan products and consequently lowering the glucan content [32]. Accordingly, the optimal molar ratio of $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) selected in this study was 1:1:1.

Pretreatment duration can considerably impact the chemical composition of WS. $TDES_{Gly}$ ChCl:*p*-TsOH:glycerol (molar ratio = 1:1:1, solid-liquid ratio = 1:10 wt/wt) was adopted to pretreat WS for varying durations (30 min to 75 min) at 80 °C (Fig. 3a). Upon undergoing the $TDES_{Gly}$-pretreatment from 30 min to 45 min, the glucan content in the $TDES_{Gly}$-treated WS rose from 66.2 % to 70.8 %, while the lignin content dropped from 17.2 % to 4.8 % (Fig. 3a). In 45 min, the glucan recovery reached 68.9 %, and the lignin removal was as high as 88.5 %, with xylan being nearly entirely eliminated. Further extension of the $TDES_{Gly}$-pretreatment duration didn't result in remarkable improvements in eliminating xylan and lignin. Under certain performance conditions, reactions such as lignin depolymerization and xylan hydrolysis might reach chemical equilibrium. Prolonging the operation duration no longer favored to improve the removal efficiency of these components [33]. Consequently, the optimal $TDES_{Gly}$-pretreatment time was determined to be 45 min.

$TDES_{Gly}$-pretreatment temperature may also have a notable impact on the chemical composition of WS. $TDES_{Gly}$ ChCl:*p*-TsOH:glycerol (molar ratio = 1:1:1, solid-liquid ratio = 1:10 wt/wt) was adopted to pretreat WS for 45 min at different temperatures (40 °C to 120 °C). As evidenced in Fig. 4a, increasing the $TDES_{Gly}$-treatment temperature

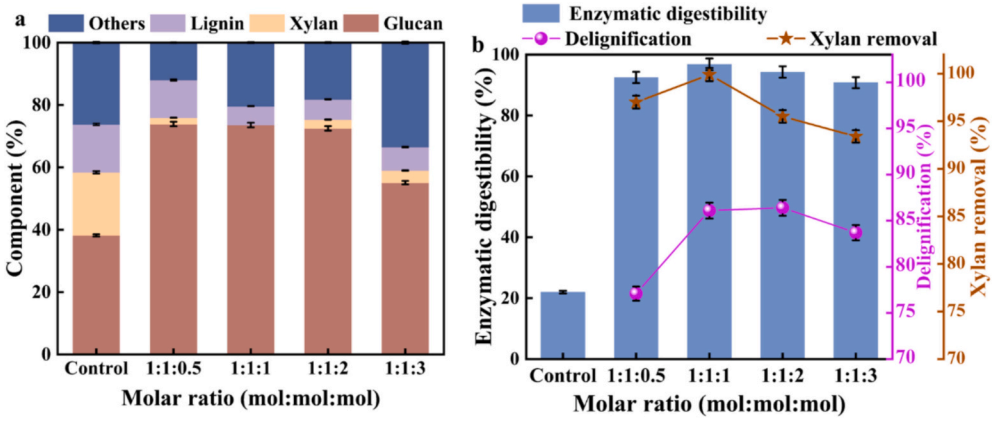


Fig. 2. Compositional changes of WS treated with different $TDES_{Gly}$ [molar ratio ranging from 1:1:0.5 to 1:1:3, 80 °C, 60 min, and the solid/liquid ratio = 1:10 wt/wt] (a); The impact of different $TDES_{Gly}$ on enzymatic digestibility, xylan removal and delignification [molar ratio ranging from 1:1:0.5 to 1:1:3, 80 °C, 60 min, and the solid/liquid ratio = 1:10 wt/wt] (b). Control: Untreated WS.

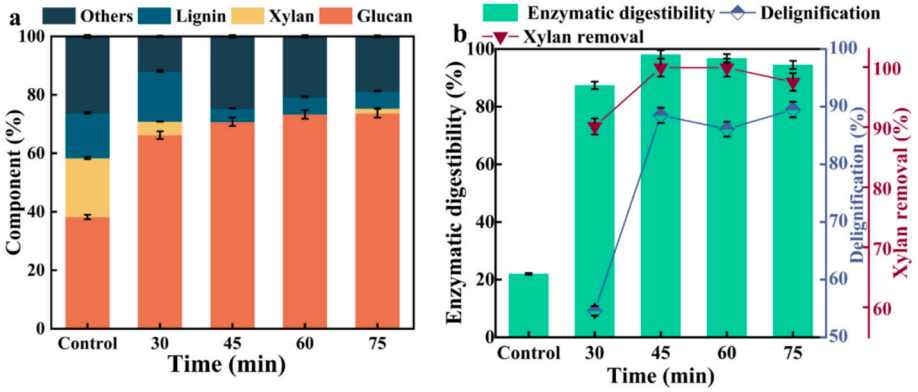


Fig. 3. Compositional alterations of WS treated under different $TDES_{Gly}$-pretreatment duration ranging from 30 to 75 min [molar ratio = 1:1:1, 80 °C, and the solid/liquid ratio = 1:10 wt/wt] (a); Different $TDES_{Gly}$-pretreatment duration ranging from 30 to 75 min on influencing enzymatic digestibility, xylan removal, and delignification. [molar ratio = 1:1:1, 80 °C, and the solid/liquid ratio = 1:10 wt/wt] (b). Control: Untreated WS.

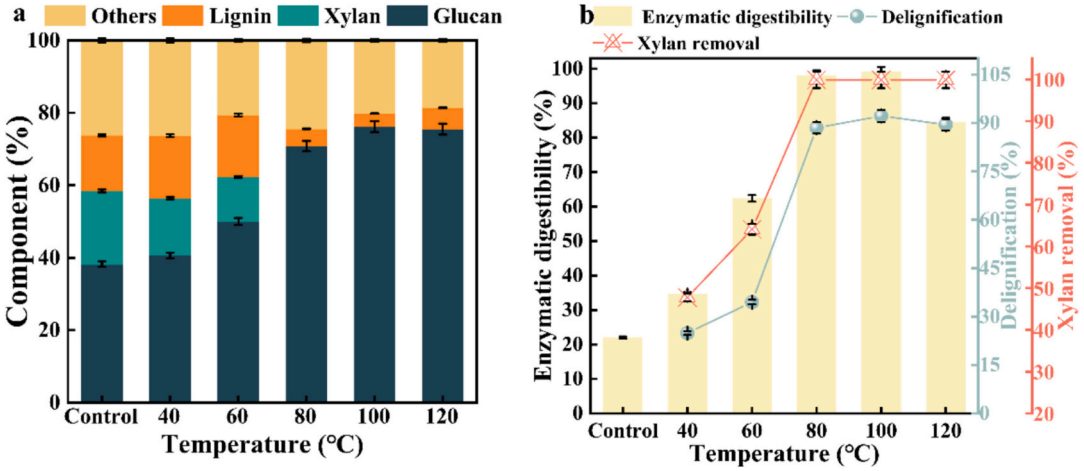


Fig. 4. Compositional alterations of WS treated with different performance temperature ranging from 40 to 120 °C [molar ratio = 1:1:1, 60 min, and the solid/liquid ratio = 1:10 wt/wt] (a); The impact of different operation temperature ranging from 40 to 120 °C on enzymatic digestibility, xylan removal and delignification [molar ratio = 1:1:1, 60 min, and the solid/liquid ratio = 1:10 wt/wt] (b). Control: Untreated WS.

from 40 °C to 80 °C led to a remarkable increase in glucan content from 40.6 % to 70.8 %. Concurrently, the lignin separation ratio elevated from 24.9 % to 88.5 %, and the xylan removal rose from 47.8 % to 99.9 % (Fig. 4b). This improvement could be explained by the fact that elevating temperatures might facilitate to cleave inter-unit bonds within lignin structures and promote the hydrolysis of glycosidic bonds in xylan, consequently improving their removal efficiency [34]. As lignin and xylan were eliminated, the glucan previously encapsulated or bound

to them becomes liberated, leading to an increase in its relative content [30]. Under the elevated $TDES_{Gly}$-treatment temperature, the lignin content in the residual WS solid phase began to rise, and the lignin removal was under plateaus. At excessively high treatment temperatures, partially depolymerized lignin fragments might undergo condensation reactions and re-precipitate into the solid residue. This phenomenon not only increased the lignin content in the solid residue [35], but also hampered further interaction between the $TDES_{Gly}$ ChCl:*p*-TsOH:glycerol system and lignin, resulting in minimal improvement in lignin removal efficiency [36]. Accordingly, 80 °C was selected as the optimal ChCl:*p*-TsOH:glycerol treatment temperature.

The relationships between the combined hydrolysis factor (CHF) and residual xylan (XR), as well as between the combined delignification factor (CDF) and residual lignin (LR), can be established and fitted [37,38]. As evidenced in Fig. 5, both the CHF–XR relationship (R^2 = 0.96) and the CDF–LR relationship (R^2 = 0.99) exhibited strong correlation coefficients. As the CHF value increased from 1.7 to 10, the residual xylan content decreased rapidly. A further increase in CHF from 10.0 to 35.0 resulted in a more gradual decline in residual xylan (Fig. 5a). A similar trend was observed for residual lignin content (Fig. 5b). This behavior may be attributed to the initial rapid dissolution of readily soluble free xylan, followed by the slower removal of unreacted xylan, which is tightly associated with cellulose and lignin, thereby requiring the overcoming of greater steric hindrance. Lignin removal is likely influenced by its polymeric structure—loosely bound superficial lignin is extracted first, while the internal dense cross-linked structure impedes further reaction. For the ChCl/*p*-TsOH/Gly pretreatment system, when the CHF exceeds 10.0 or the CDF exceeds 127.3, high removal rates of xylan (99.9 %) and lignin (88.5 %) can be achieved, indicating optimal pretreatment performance. Accordingly, XR and LR could be effectually predicted using CHF and CDF, respectively. CHF and CDF can serve as reliable indicators of $TDES_{Gly}$ pretreatment severity. With increasing $TDES_{Gly}$-treatment intensity, substantial amounts of xylan and lignin were considerably eliminated [39]. Accurate prediction of lignin and xylan removal can help prevent resource waste during scale-up processes and provide robust support for the large-scale industrial utilization of biomass resources in the future.

3.3. Enzymatic digestion

Using $TDES_{Gly}$ ChCl:*p*-TsOH:glycerol effectually disrupted the compact structure of biomass and separates lignin and xylan that are tightly bound to cellulose, consequently facilitating subsequent enzymolysis. Difference from untreated WS, the enzymatic digestibility of DES-pretreated WS was considerably enhanced. Under identical treatment conditions (80 °C, 60 min), the enzymolysis efficiency of WS after $TDES_{Gly}$ treatment reached 96.8 %, which was 4.4-folds higher than that of raw WS (22.0 %) (Fig. 1). In conjunction with the compositional analysis results, this improvement can be attributed to the disruption of the complex lignocellulosic matrix by $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) treatment, eliminating a greater amount of lignin and exposing more cellulose surface area and enzyme-accessible sites [40,41]. Furthermore, supplementing glycerol (Gly) enables hydroxyl groups to interact with cellulose, forming a more stable structure that preserves cellulose content. This increases the substrate concentration available for enzymolysis and further improves the overall enzymolysis rate [41].

The molar ratio of the ternary DES components is a critical factor impacting enzymatic digestibility. Upon adjusting the ChCl:*p*-TsOH:glycerol molar ratio from 1:1:0.5 to 1:1:1 (solid-liquid ratio = 1:10 wt/wt, 80 °C, 60 min), the enzymatic digestion rate was improved from 92.5 % to 96.8 %. Further increasing the proportion of glycerol led to a diminishment in enzymolysis efficiency (Fig. 2b). This phenomenon may be attributed to the excessive hydroxyl groups from glycerol competing with the hydrogen-bond acceptor sites originally present in the ChCl-based DES system. As a result, the system's capacity to accept protons and form hydrogen-bonds was diminished, weakening its ability to eliminate lignin and xylan. Pretreatment duration also plays a significant role in determining enzymatic digestibility. Extending the performance duration from 30 min to 45 min resulted in an increased enzymatic digestion from 87.4 % to 98.1 %. Prolonging the time to 75 min caused a decrease in hydrolysis efficiency to 94.5 % (Fig. 3b). Both insufficient and excessive operation durations are detrimental to the efficient removal of lignin and xylan, which in turn impacts the accessibility of cellulose to enzymes during the hydrolysis [40]. Operation temperature is another key parameter impacting enzymatic digestion performance (Fig. 4b). As the performance temperature was raised from 40 °C to 80 °C, the enzymatic digestion rapidly increased from 34.7 % to 98.1 %. Beyond 80 °C, the enhancement slowed. At 120 °C, the digestion rate even declined to 84.5 %. Higher performance temperatures might enhance the structural disruption of lignocellulosic materials, exposing more cellulose to enzymatic attack and thus improving hydrolysis efficiency [42]. While excessively high operation temperatures can lead to the generation of a more complex and compact structure that re-encapsulates cellulose, limiting enzyme access and consequently lowering the enzymatic digestion rate [40,43].

Furthermore, this study employed response surface methodology (RSM) to design an optimized experimental strategy. The independent variables included: A—performance temperature (60–100 °C), B—operation time (30–60 min), and C—the molar ratio of $TDES_{Gly}$

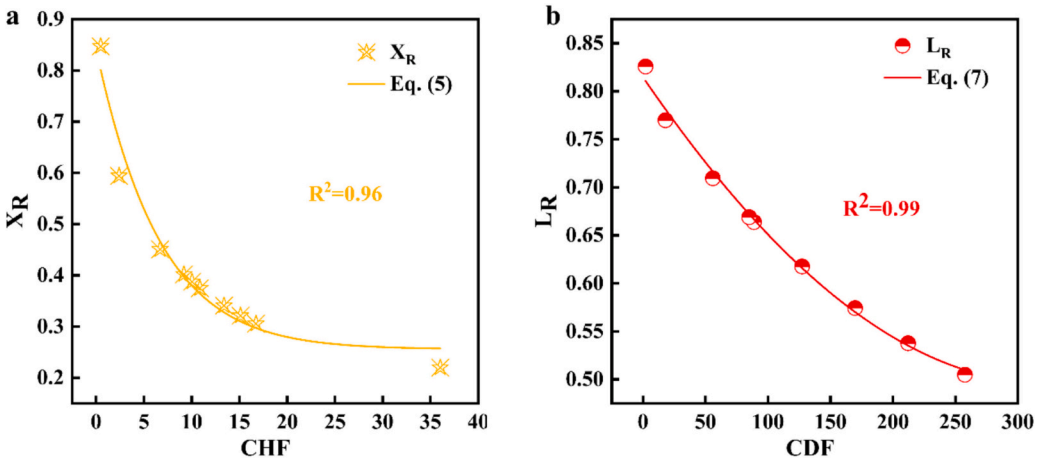


Fig. 5. Experimental measured xylan (a) and lignin (b) dissolution under various fractionation severities along with fittings of the data sets to reaction kinetic-based severity factors.

(1:1:0 to 1:1:2, mol/mol/mol). A total of 17 performance experiments were conducted using $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol), each repeated three times. The average enzymatic digestion rate was calculated for each condition, and corresponding compositional analyses were performed. The performance temperature and duration of $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) could considerably impact enzymatic digestibility (Fig. 6a). As the performance duration was held constant, an initial increase in temperature led to a remarkable enhancement in saccharification efficiency. However, this efficiency subsequently declined with further temperature elevation. This phenomenon can be attributed to the fact that higher temperatures reduce the molecular activation energy and accelerate the hydrolysis reaction [44]. Under the action of $TDES_{Gly}$, a substantial portion of lignin and hemicellulose can be eliminated consequently exposing accessible cellulose structures and improving biomass digestibility. However, excessively high temperatures can induce side-reactions or cause partial cellulose degradation, favoring the generation of numerous enzyme inhibitors [40,43]. Elevating performance temperatures might favor the production of pseudo-lignin, which is prone to adsorbing on the biomass surface [45] and consequently hampering enzymatic digestibility [45].

Based on the two-factor analysis of performance temperature, performance duration, and DES molar ratio (Fig. 6b & c), it is evident that temperature or duration was held constant, and the saccharification efficiency of WS gradually increased upon raising the glycerol molar ratio in DES from 1:1:0 to 1:1:1. This trend may be attributed to the hydrogen-bond interactions between glycerol and the other components, ChCl and *p*-TsOH, which together form a complex and distinctive hydrogen-bond network [46]. During the DES treatment process, the hydroxyl groups of glycerol can form hydrogen-bonds with phenolic hydroxyl groups, carbonyl groups, and other functional groups present in lignin molecules, weakening both intramolecular and intermolecular forces within the lignin structure, forming a more relaxed configuration to facilitate its detachment from the WS cell wall, and consequently promoting its solubility in DES [47,48]. Furthermore, supplementing glycerol (as the third component) helps modulate the viscosity of DES, promoting better penetration of the solvent into the cell wall of WS. This allows for more effective contact between DES and the lignin and xylan components, consequently augmenting the accessibility of cellulose to enzymes and enhancing the efficiency of enzymolysis [41]. As the glycerol component molar ratio in $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) exceeded 1:1:1, the enzymatic digestibility began to diminish. A high molar ratio of glycerol in DES may create a more severe pretreatment environment, potentially leading to the generation of inhibitory by-products that negatively impact enzymolysis and accumulate on the surface of the biomass [49,50]. In a concise summary, the key parameters of $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) treatment were optimized using response surface methodology. Under the optimal conditions—namely, a DES ChCl:*p*-TsOH:glycerol molar ratio of 1:1:1.3, a performance time of 55 min, and a performance temperature of 80 °C—the highest enzymatic digestibility of 98.5 % was achieved.

3.4. Physicochemical characterization of WS before and after DES treatment

3.4.1. Cellulose accessibility

Cellulose accessibility is widely recognized as a critical factor impacting the efficiency of enzymolysis of lignocellulosic substrates [14]. In this study, the initial cellulose accessibility of raw WS was measured at 306.6 mg/g. As the molar ratio of choline chloride (ChCl), *p*-toluene sulfonic acid (*p*-TsOH), and glycerol in $TDES_{Gly}$ increased from 1:1:0.5 to 1:1:1 (solid-liquid ratio = 1:10 wt/wt, 80 °C, 60 min), the cellulose accessibility was improved from 563.3 mg/g to 664.7 mg/g. Under the optimized pretreatment conditions of $TDES_{Gly}$ (molar ratio = 1:1:1.3, 80 °C, 55 min), the cellulose accessibility reached 691.5 mg/g. It demonstrates that the composition ratio and processing conditions have a significant impact on cellulose accessibility.

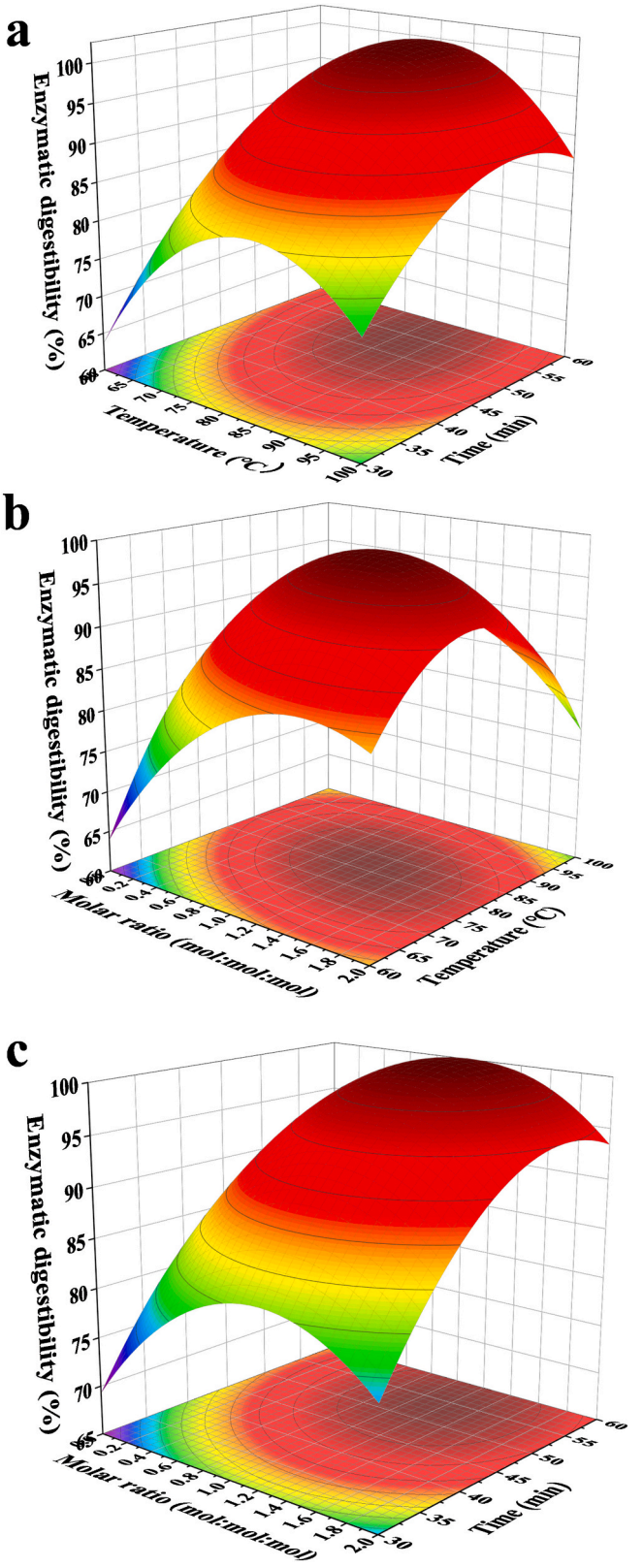


Fig. 6. Response Surface modeling of the effects of pretreatment temperature (60–100 °C) and time (30–60 min) on enzymatic digestibility (a); Impacts of performance temperature (60–100 °C) and $TDES_{Gly}$ molar ratio (1:1:0–1:1:2) on enzymatic digestibility (b); Impacts of performance duration (30–60 min) and $TDES_{Gly}$ molar ratio (1:1:0–1:1:2) on enzymatic digestibility (c).

It has been reported that lignin and hemicellulose form a protective barrier around cellulose in the plant cell wall, which hampers the adsorption of cellulase onto cellulose and consequently limits enzymolysis efficiency [51]. During $TDES_{Gly}$ treatment, the effective removal of lignin and xylan significantly can increase the exposed surface area of cellulose, consequently enhancing enzyme accessibility [52]. After treatment with $TDES_{Gly}$ under a molar ratio of 1:1:1, a significant decrease in the contents of lignin and xylan was observed, which is very likely the structural basis for the improved accessibility of cellulose. To further estimate this relationship, the lignin removal rate exhibits a strong positive correlation with cellulose accessibility ($R^2 = 0.90$), and the xylan removal rate demonstrates a similarly high correlation with cellulose accessibility ($R^2 = 0.89$), revealing a strong positive correlation, as illustrated in Fig. 7. Additionally, a linear regression analysis was acquired between enzymatic digestibility and cellulose accessibility [$Y_{(accessibility)} = 5.31 \times X_{(enzymatic\ digestibility)} + 117.18$, $R^2 = 0.89$], indicating a significant positive association (Fig. 9b). These results manifested that $TDES_{Gly}$ pretreatment significantly enhanced cellulose accessibility by efficiently removing lignin and xylan, thereby ultimately improving enzymatic digestibility.

3.4.2. Lignin surface area

A larger lignin surface area indicates that a greater amount of lignin interferes with the interaction between cellulase and cellulose [52]. In this study, the adsorption effect of calcein B dye was utilized to determine the specific surface area of lignin, and the relationships between the lignin surface area, the removal rate of xylan, and deoligomerization were investigated. Following treatment with $TDES_{Gly}$ at 80 °C for 45 min, the surface area of lignin decreased significantly to 134.6 m^2/g, compared to the initial surface area of 258.6 m^2/g in untreated lignin fibers. The pre-treatment with $TDES_{Gly}$ further reduced the physical barrier formed by residual lignin on the surface of lignin fibers, thereby enhancing the dissolution of lignin by TDES. Additionally, the lignin structure became more porous, which may have facilitated the exposure of more xylan polysaccharide reaction sites in the pretreatment solution, leading to an increased xylan polysaccharide removal rate. Regression analysis revealed that lignin surface area exhibited a strong negative correlation with both the lignin removal ($R^2 = 0.88$) and the xylan removal ($R^2 = 0.94$), as indicated in Fig. 8. As enzymolysis efficiency increased, the lignin surface area demonstrated a decreasing trend, indicating a significant inverse relationship between enzymatic digestibility and lignin surface area ($R^2 = 0.95$) (Fig. 9b). These findings clearly demonstrated that ChCl:*p*-TsOH:glycerol treatment effectually diminished lignin surface area, promoted further

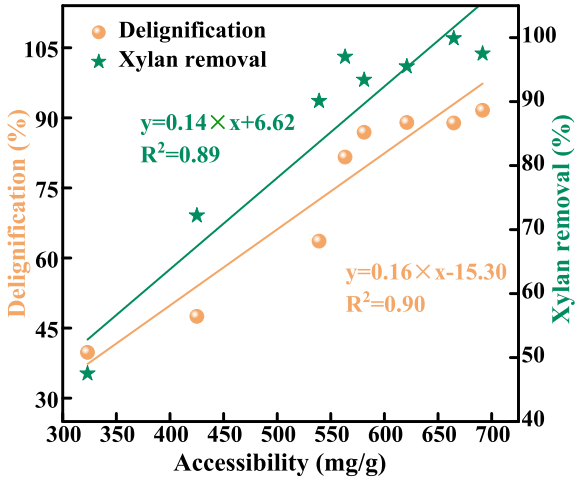


Fig. 7. The linear fitting about accessibility with xylan removal and delignification.

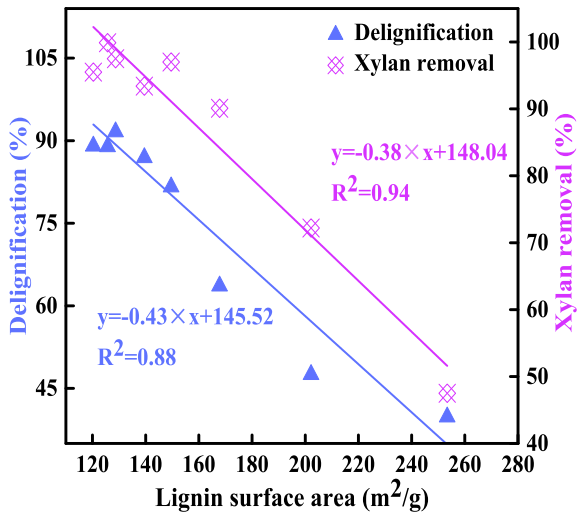


Fig. 8. The linear fitting about surface area of lignin with xylan removal and delignification.

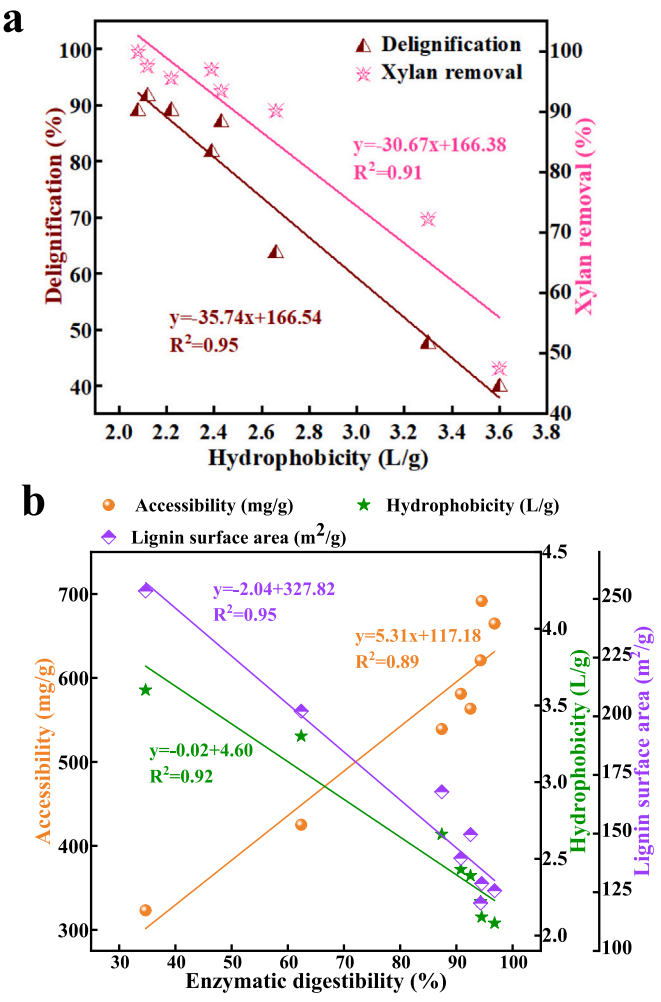


Fig. 9. Linear fitting about hydrophobicity with xylan removal and delignification (a); Linear fitting about enzymatic digestibility to cellulose accessibility, lignin surface area and hydrophobicity (b).

dissolution of lignin in WS, and consequently improved enzymatic digestion efficiency. However, excessively intense pretreatment may compromise the structural integrity of cellulose and hemicellulose, thereby impeding the efficient acquisition of fermentable sugars.

3.4.3. Hydrophobicity

Lignin contains abundant hydrophobic groups, such as aromatic rings and aliphatic side chains. These groups can specifically interact with the hydrophobic structure of cellulase through hydrophobic forces, causing cellulase to adhere to lignin and thereby hindering the enzymatic hydrolysis process [53]. In this work, the hydrophobicity of lignin in raw WS was measured at 3.9 L/g, whereas that of WS following binary DES (ChCl:*p*-TsOH) treatment (80 °C, 60 min) decreased to 3.2 L/g. This is because ChCl:*p*-TsOH is capable of cleaving the ether bonds (e.g., the β-O-4 bond) between lignin's structural units, oxidizing the hydrophobic aromatic rings, and simultaneously reducing both the size and density of lignin's hydrophobic domains. After $TDES_{Gly}$ (ChCl:*p*-TsOH:glycerol) treatment, which resulted in the remarkable removal of lignin, the hydrophobicity further declined from 3.2 L/g to 2.6 L/g. Glycerol, as a hydrogen bond donor/acceptor cosolvent, promotes the cleavage of lignin by forming hydrogen bonds with the hydroxyl and carboxyl groups of lignin, while spatially inhibiting the re-aggregation of lignin fragments, thereby further reducing hydrophobicity. As evidenced in Fig. 9a, a strong negative correlation was observed between lignin hydrophobicity and both the lignin removal ($R^2 = 0.95$) and the xylan removal ($R^2 = 0.91$). The diminished lignin content led to a decrease in hydrophobic components within the DES-treated WS, consequently lowering its overall hydrophobicity. Moreover, as lignin and xylan were progressively eliminated, more cellulose becomes exposed. The hydroxyl groups present in cellulose can then interact more readily with H_2O molecules, further contributing to the reduction in lignin hydrophobicity in the pretreated WS [54]. The hydroxyl groups of cellulose establish extensive hydrogen bonding networks with water molecules, leading to the formation of a hydration layer that physically envelops the cellulose fibers and consequently repels the hydrophobic domains of lignin. Fig. 9b indicates a significant negative correlation between lignin hydrophobicity and enzymatic digestibility ($R^2 = 0.92$). As ChCl:*p*-TsOH:glycerol treatment progressed, it cleaved the ether bonds within large lignin molecules, reduced the hydrophobic aromatic domains, and introduced additional polar functional groups, such as hydroxyl and carboxyl groups, onto the lignin fragments. These structural modifications not only decreased the number of irreversible interaction sites between lignin and cellulase but also reduced their binding affinity. Specifically, the reduction of large aromatic clusters diminished the π-π stacking interactions with the aromatic residues in cellulase, whereas the increased presence of polar groups on lignin enhanced its hydration capacity, thereby further weakened hydrophobic attractions to cellulase. Collectively, these alterations indicated a substantial decrease in lignin's hydrophobicity, which facilitated more efficient enzymatic hydrolysis [55].

3.5. Microstructural analysis of WS before and after $TDES_{Gly}$ treatment

3.5.1. XRD

The crystallinity of the raw material, as well as the solid residues after DES-ChCl and $TDES_{Gly}$ treatment, was analyzed using X-ray diffraction (XRD) [56]. As depicted in Fig. 10, the crystallinity index (*CrI*) of the raw material was 36.2 %, whereas the *CrI* values for DES-ChCl-pretreated WS and $TDES_{Gly}$-pretreated WS were significantly higher at 44.2 % and 41.8 %, respectively. This enhancement in crystallinity can be attributed to the effective dissolution or degradation of lignin and hemicellulose during pretreatment [57]. However, the crystallinity of $TDES_{Gly}$-pretreated WS was slightly lower compared to that of DES-ChCl-pretreated WS. This may be due to the presence of glycerol, which can partially protect the amorphous regions of cellulose from structural disruption and preserve their original state, thereby leading to a slight reduction in overall crystallinity [58]. During the enzymolysis, cellulase primarily targets the amorphous regions of cellulose [59]. Accordingly, supplementing glycerol in DES for WS pretreatment could promote enzyme accessibility to these amorphous regions, ultimately improving the enzymatic digestibility.

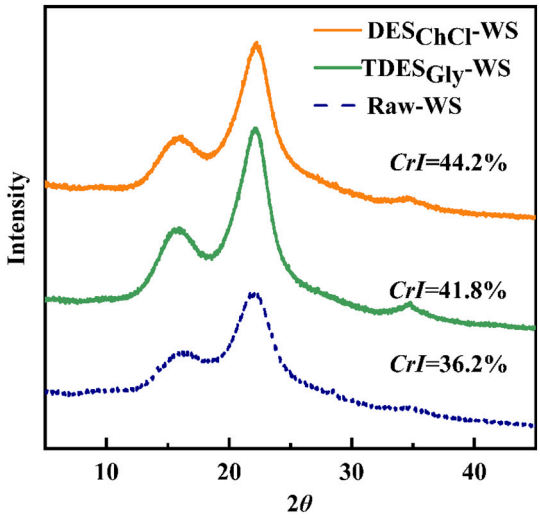


Fig. 10. XRD of raw and pretreated WS.

3.5.2. FT-IR

Fig. 11 presents the Fourier-transform infrared (FT-IR) spectra of raw WS and WS after DES-ChCl and $TDES_{Gly}$ pretreatment. The absorption peak around 3420 cm^{-1} is mainly attributed to the O—H stretching vibration of the hydroxyl functional groups in cellulose and hemicellulose [60]. Difference from raw WS, the intensity of the peak near 3420 cm^{-1} in the solid residues after DES treatment was diminished, which can be attributed to the removal of hemicellulose and partial degradation of cellulose, resulting in a decrease in the number of hydroxyl groups and consequently a weakening of the O—H stretching signal [61]. Peaks observed near 806 cm^{-1} and 2925 cm^{-1} may be related to C—H vibrations in lignin and carbohydrates [62]. The partial removal or structural modification of lignin during DES treatment leads to a reduction in the number of C—H bonds, consequently weakening the intensity of the peak near 2925 cm^{-1} [30]. The absorption peak around 1621 cm^{-1}, corresponding to the C═C stretching vibration in the aromatic rings of lignin, has also weakened in intensity, indicating the effective removal of lignin [63]. The absorption band at approximately 1108 cm^{-1} is believed to originate from the C-O-C stretching vibration in lignin or hemicellulose. The attenuation of this peak suggests that chemical transformations such as demethylation and cleavage of ether bonds occurred during the DES-treatment process [64]. The changes in these characteristic absorption peaks are directly attributed to the removal and structural modification of lignin and carbohydrates by the TDES-based pretreatment.

3.5.3. SEM

The scanning electron microscopy (SEM) images presented in Fig. S1 (see Supplementary files) illustrate the surface morphology of untreated and $TDES_{Gly}$-pretreated WS samples. The surface of untreated WS appears dense and smooth, indicating limited accessibility for cellulase for enzymolysis of WS (Fig. S1a, see Supplementary files). In contrast, after $TDES_{Gly}$ treatment, the WS surface exhibits fragmentation, visible cracks, and a more porous structure, resulting in increased surface roughness, specific surface area and accessibility (Fig. S1b, see Supplementary files). These structural modifications, primarily attributed to the removal of lignin, may enhance enzyme accessibility and facilitate better interaction between cellulase and cellulose, consequently

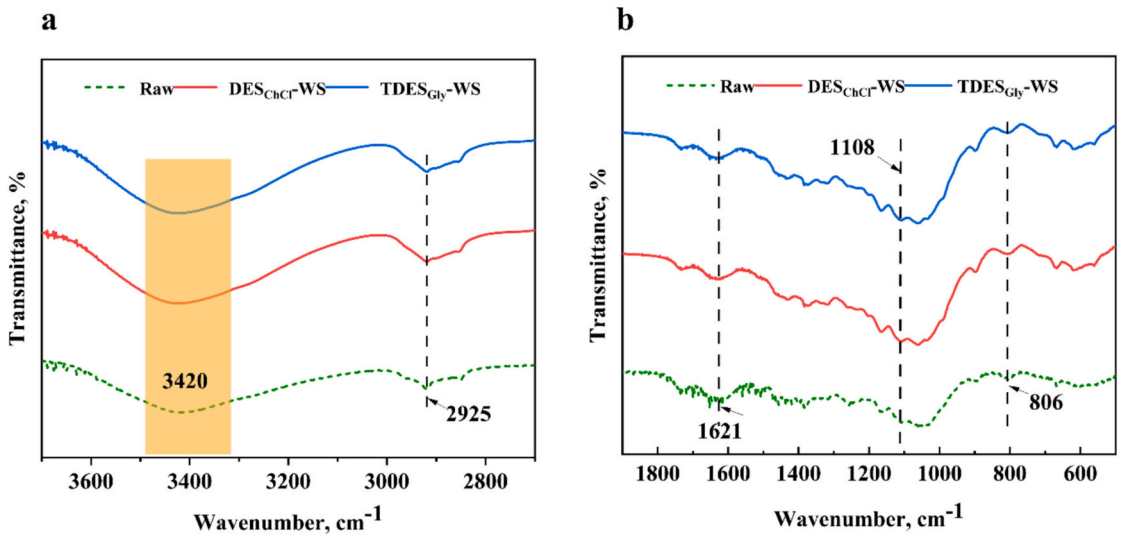


Fig. 11. FT-IR of raw and pretreated WS.

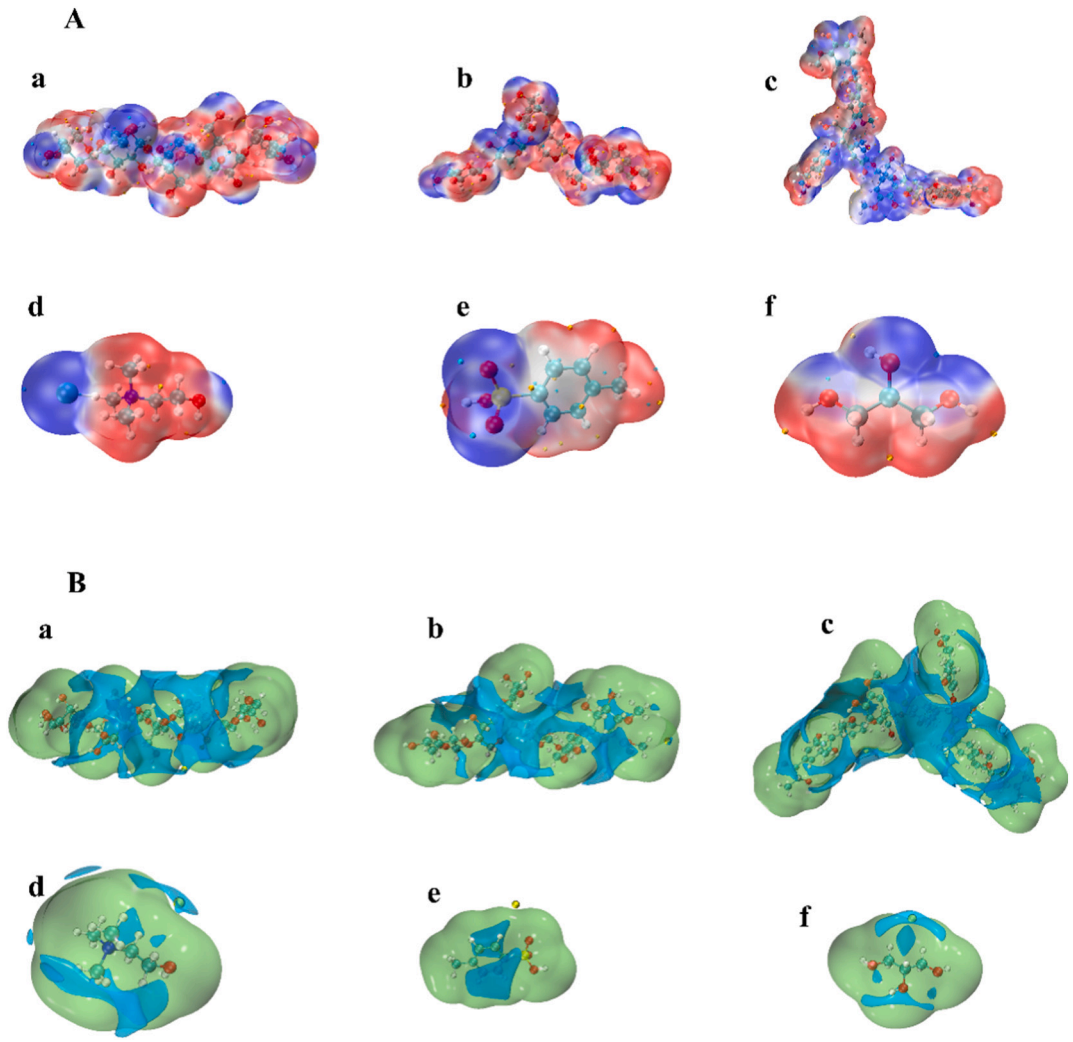


Fig. 12. Electrostatic potential (A) and van der Waals forces (B) of cellulose (a), hemicellulose (b), lignin (c), ChCl (d), *p*-TsOH (e), and Gly (f).

improving the efficiency of enzymolysis.

3.6. Molecular dynamics simulation of lignocellulosic model in $TDES_{Gly}$

Understanding the molecular-level interactions of lignocellulosic biomass can considerably advance the biomass pretreatment [65]. The electrostatic potential and van der Waals force distribution maps of the three components of $TDES_{Gly}$ and three representative lignocellulosic model compounds were obtained (Fig. 12), providing valuable insights into the intermolecular mechanisms involved. Specifically, the hydroxyl oxygen atoms in lignocellulose exhibit negative electrostatic potential (ESP) values, whereas the hydroxyl hydrogen atoms display positive ESP values [66]. ESP maps (Fig. 12) provide detailed insights into the charge distribution within the components of WS. The charge distribution of the $TDES_{Gly}$ components demonstrates strong interactions with that of the biomass constituents. Notably, *p*-TsOH and Gly in DES introduced a pronounced negative electrostatic potential into the system, considerably disrupting the hydrogen-bond network present in lignin and hemicellulose. This electrostatic potential directly impacts the hydrogen-bonds within the lignin and hemicellulose structures (Fig. 12A). The introduction of the DES's electrostatic potential greatly interferes with the formation and stability of these hydrogen-bonds through electrostatic repulsion, consequently weakening the intermolecular interactions between lignin and hemicellulose. Van der Waals

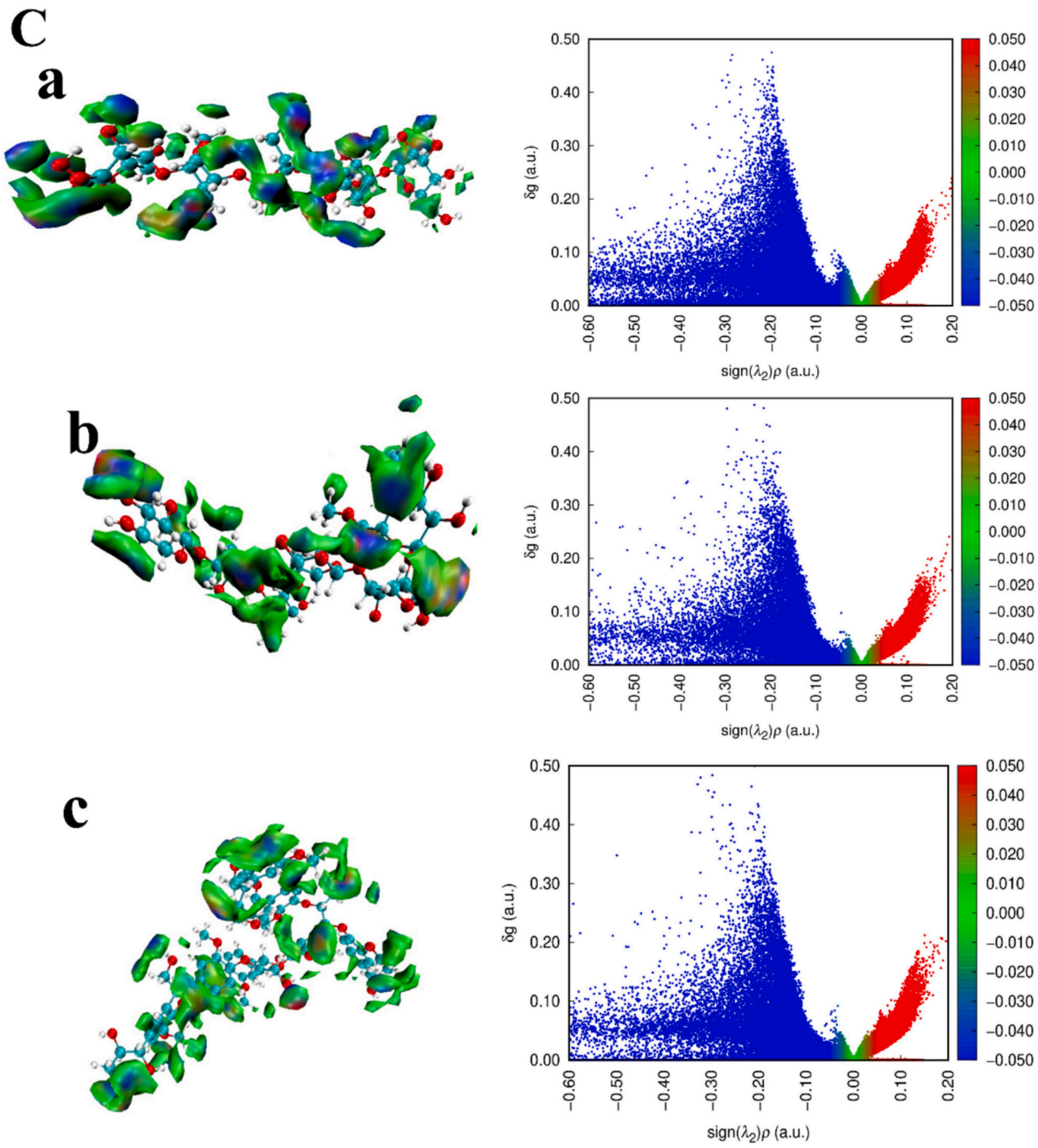


Fig. 13. Isosurface images and scatter plots of the independent gradient model (IGM) for the cellulose model (a), hemicellulose model (b), and lignin model (c) during ChCl/*p*-TsOH/Gly treatment.

forces also play a significant role in the interactions between DES and biomass components, particularly in facilitating the association of non-polar compounds [67]. In the van der Waals potential energy map, the blue regions indicate areas where van der Waals interactions are predominantly attractive (Fig. 12B). Both sides of the ChCl surface are enveloped by blue regions, suggesting that ChCl exhibits strong dispersion interactions with non-polar molecules present in biomass. Fig. 12Bc manifests that *p*-TsOH possesses notable polarity and solvation capability, enabling it to effectually eliminate non-polar components, such as lignin, from the lignocellulosic structure. Moreover, in the lignocellulose model, the lignin model is extensively surrounded by blue regions, revealing that the dispersion interactions between the lignin model and the solvent are the most pronounced.

The Independent Gradient Model (IGM) is known as a computational analytical tool than can be adopted to identify strong non-covalent interactions among atoms or molecules [68,69]. This work comprehensively investigated the interaction actions between the lignocellulosic components (cellulose, hemicellulose, and lignin) and the ChCl/*p*-TsOH/Gly through the molecular dynamics (MD) simulations and independent gradient model (IGM) analysis, revealing that hydrogen-bonding represents a key non-covalent interaction between model components and $TDES_{Gly}$ molecules, primarily occurring between hydrogen atoms in strong polar bonds and lone pair electron acceptors (Fig. 13). Notably, hemicellulose, which is rich in hydroxyl groups, exhibits both hydrogen-bond donor and acceptor capabilities. The IGM testing (Fig. 13b) showcased a dense region of dark blue at the interface between hemicellulose and $TDES_{Gly}$, indicating the occurrence of a robust hydrogen-bond network. The isosurface visualization apparently manifested that the interaction strength between hemicellulose and $TDES_{Gly}$ via hydrogen-bonding was the highest among all components. Given that hydrogen-bonds are a primary form of electrostatic interaction, this result directly manifests that the strongest hydrogen-bonding occurs between hemicellulose and ChCl/*p*-TsOH/Gly. These strong interactions facilitate the hydrolysis of hemicellulose and greatly disrupt the covalent bonds between sugar molecules. In contrast, lignin consists of rigid aromatic rings connected by aryl ether bonds and flexible side-chains linked by carbon-carbon bonds, with a limited number of strong polar functional groups. Consequently, the interaction between lignin and $TDES_{Gly}$ is predominantly governed by van der Waals forces. IGM testing (Fig. 13c) revealed extensive green regions at the lignin–solvent interface, corresponding to van der Waals interactions within this analytical framework. This observation can be attributed to the overall low polarity and pronounced hydrophobic nature of the lignin macromolecule. Based on the electrostatic potential analysis from MD simulations and IGM investigations, the fundamental differences in the interactions between lignocellulosic components and ChCl/*p*-TsOH/Gly were elucidated at the electronic structure level, offering critical insights into the selective dissociation behavior of biomass in this DES system.

3.7. Mass flow and future prospects

Fig. S2 (see Supplementary files) presents the mass flow of $TDES_{Gly}$ pretreatment and subsequent enzymolysis of WS. The initial 100 g of raw WS contains 38.2 g of glucan, 20.2 g of xylan, 15.4 g of lignin, and 26.2 g of other components. After $TDES_{Gly}$ pretreatment under conditions of a molar ratio of 1:1:1, 80 °C for 45 min, and a solid-to-liquid ratio of 1:10 (wt/wt), the resulting solid residue (37.2 g) consists of 26.3 g of glucan, 1.8 g of lignin, and 9.1 g of other substances, with xylan being almost entirely eliminated. Enzymolysis of the pretreated WS using cellulase at a dosage of 10 FPU/g glucan yielded 28.1 g of glucose. Based on the original glucan content (38.2 g), approximately 68.9 % was retained in the solid residue after DES pretreatment. Of this retained glucan, 98.1 % of them was successfully transformed into fermentable glucose during enzymolysis. Nearly complete removal of xylan and 88.5 % of lignin separation were observed. These results indicate that $TDES_{Gly}$ treatment demonstrates highly effectual degradation of lignin and xylan. Consequently, glucan was effectually transformed into glucose. In a concise summary, $TDES_{Gly}$ treatment offers significant merits in terms of biomass component separation and enhancement of enzymolysis efficiency. It not only ensures effective retention of glucan during pretreatment but also enables its efficient conversion into fermentable glucose. This provides a high-quality substrate for biofuel production, biobased chemicals.

The $TDES_{Gly}$ ChCl:*p*-TsOH:Gly treatment strategy enhances enzymatic digestibility by efficiently disrupting WS structure (Fig. 14). In ChCl:*p*-TsOH:Gly, the *p*-TsOH component, as a strong acidic component, may create an acidic environment in which β-O-4 bonds within lignin and glycosidic bonds in hemicellulose are readily cleaved under acidic conditions, thereby promoting lignin depolymerization and hemicellulose degradation [70]. Concurrently, the ChCl and Gly components in ChCl:*p*-TsOH:Gly can disrupt the hydrogen-bond network among cellulose, hemicellulose, and lignin in the WS cell wall through hydrogen-bonding interactions. This process may reduce the crystallinity of the biomass and diminish its resistance to enzymatic degradation. Incorporating glycerol considerably increases the glucan retention rate, ultimately enhancing enzymatic digestibility [58]. Compared with binary DES systems, $TDES_{Gly}$ achieved higher removal efficacies for both lignin and xylan within a shorter reaction duration, revealing that supplementing glycerol remarkably improves the pretreatment performance of DES. Glycerol can enhance lignin dissolution and acid hydrolysis of hemicellulose by modulating the acidity and hydrogen-bond network of the system [31]. Moreover, glycerol may impact the physicochemical properties of DES, such as viscosity and polarity, consequently facilitating greater exposure of cellulose during pretreatment and further improving enzymatic digestibility [46]. A series of characterizations indicated that significant changes occurred in WS after pretreatment with ChCl:*p*-TsOH:Gly compared with the untreated WS. The strong acidic group of *p*-TsOH disrupted the compact cellulose structure. The cellulose accessibility of the treated WS reached 691.5 mg dye/g cellulose, while that of the untreated WS was 306.6 mg/g. Compared with untreated WS (3.9 g/L), the hydrophobicity of the treated WS decreased to 2.6 g/L. This reduction in lignin hydrophobicity partially mitigated the inhibitory effect of lignin on cellulase hydrolysis, thereby improving the enzymatic hydrolysis efficiency. Additionally, the surface area of lignin decreased after pretreatment. These structural and chemical modifications provide a plausible explanation for the enhancing effect of ChCl:*p*-TsOH:Gly on the production of reducing sugars from WS.These results highlight the promising application prospects of ChCl:*p*-TsOH:Gly treatment for advancing sustainable lignocellulosic biorefining, particularly through improved efficiency and reduced operational costs. In a concise summary, the $TDES_{Gly}$-treatment system not only enhances the removal efficiency of lignin and xylan but also considerably increases cellulose accessibility, leading to the improved enzymolysis performance. This renders a more effective strategy for valorization of biomass resources.

Lignin itself holds considerable potential as a key precursor for manufacturing high-value materials. While the ChCl:*p*-TsOH:Gly treatment system demonstrates effective component separation, it may face challenges in fully deconstructing the macromolecular network structure of lignin. Current research on the biorefining process based on $TDES_{Gly}$ has not adequately addressed the acidic hydrolysis behavior and mechanism of WS mediated by this solvent system. On one hand, the dynamic interaction between the acidic sites within the $TDES_{Gly}$ system and the complex structural units of WS has not been systematically investigated, which limits the ability to clearly elucidate the degradation sequence of individual components and the bond cleavage patterns during the hydrolysis process. On the other hand, there remains a lack of quantitative analysis regarding the structure-activity relationship between key hydrolysis parameters and product selectivity, particularly concerning the formation of inhibitory by-products such as 5-hydroxymethylfurfural (HMF) and furfural. Accordingly, promoting the effectual depolymerization of lignin and improving its transformation into

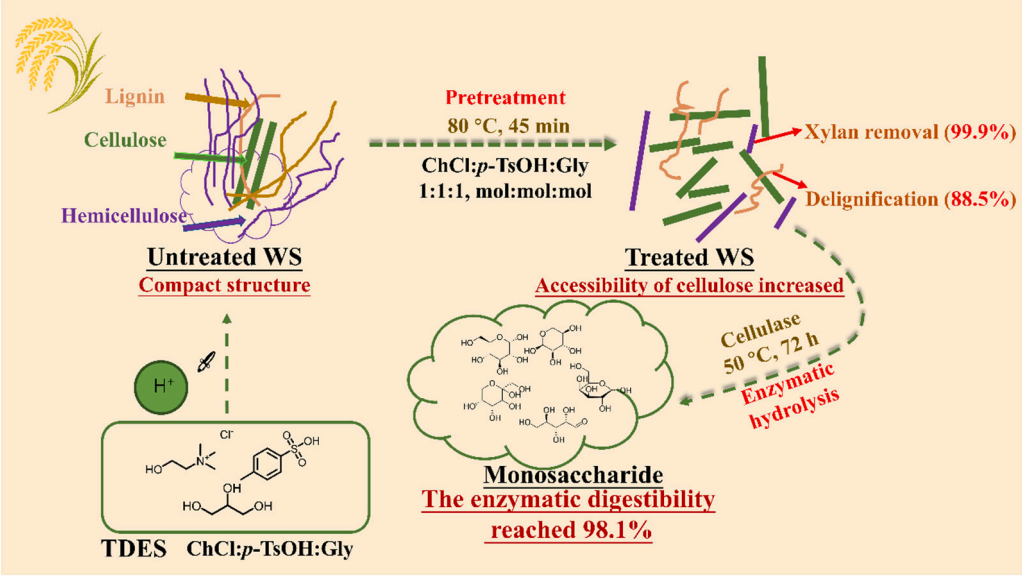


Fig. 14. $TDES_{Gly}$ pretreatment and enzymolysis of RS.

functional materials during the bio-refining process based on $TDES_{Gly}$ treatment warrant in-depth investigation. Future efforts will emphasize the investigation of high-value utilization pathways for lignin, with a focus on its effectual transformation into platform chemicals. This will render a foundation for the development of downstream industries, including bioplastics and food additives. Future studies will explore the application potential of the ChCl:*p*-TsOH:Gly treatment technology with the objective of economically and efficiently converting cellulose, hemicellulose, and lignin into market-competitive biofuels and bio-based materials.

4. Conclusions

By analyzing the alterations in chemical composition and enzymolysis efficiency of WS before and after three component DES treatment, $TDES_{Gly}$ ChCl:*p*-TsOH:Gly (1:1:1) was screened for treating WS (80 °C, 45 min) to remove xyan, eliminate lignin, elevating subsequent enzymatic saccharification efficiency. Afterwards, the CDF and CHF exhibited significant predictive capability for delignification and xylan separation during the ChCl:*p*-TsOH:Gly treatment process. This predictive power provides a quantitative basis for optimizing DES pretreatment protocols. The raw materials and ChCl:*p*-TsOH:Gly-treated WS were characterized using FT-IR, XRD, and SEM techniques. The observed alterations in physicochemical properties and chemical composition manifested that the ChCl:*p*-TsOH:Gly treatment considerably improved cellulose accessibility, lowered lignin hydrophobicity, and diminished lignin surface area, resulting in the elevated enzymolysis efficiency. Molecular dynamics simulations showcased that strong hydrogen-bonding and van der Waals interactions existed between ChCl:*p*-TsOH:Gly and lignin. These findings contribute to a better understanding of the mechanisms of ChCl:*p*-TsOH:Gly treatment for improving cellulase digestibility. In conclusion, this study provides a feasible and sustainable solution for biomass processing based on TDES ChCl:*p*-TsOH:Gly technology, enabling effectual treatment under relatively mild operational conditions and realizing the valorization of lignocellulosic biomass into high-value products.

CRediT authorship contribution statement

Zhuo Zhang: Writing – original draft, Software, Methodology, Investigation, Data curation. **Qianqian Gao:** Visualization, Software, Investigation, Data curation. **Yu-Cai He:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is kindly supported by the Frontier Technology Research Plan of Jiangsu Province (BF2025080).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.147895.

Data availability

Data will be made available on request.

References

[1] F. Vidal, E.R. van der Marel, R.W.F. Kerr, C. McElroy, N. Schroeder, C. Mitchell, G. Rosetto, T.T.D. Chen, R.M. Bailey, C. Hepburn, C. Redgwell, C.K. Williams, Designing a circular carbon and plastics economy for a sustainable future, Nature 626 (7997) (2024) 45–57.
[2] L. Yang, Y. Ru, S. Xu, T. Liu, L. Tan, Features correlated to improved enzymatic digestibility of corn Stover subjected to alkaline hydrogen peroxide pretreatment, Bioresour. Technol. 325 (2021) 124688.
[3] Z. Chen, Y. Wang, H. Cheng, H. Zhou, Hemicellulose degradation: an overlooked issue in acidic deep eutectic solvents pretreatment of lignocellulosic biomass, Ind. Crop. Prod. 187 (2022) 115335.
[4] Q. Gao, Z. Tang, Y.-C. He, Valorization of wheat straw through enhancement of cellulose accessibility, xylan elimination and lignin removal by choline chloride:p-toluenesulfonic acid pretreatment, Int. J. Biol. Macromol. 301 (2025) 140335.
[5] W. Xiao, R. Sun, S. Hu, C. Meng, B. Xie, M. Yi, Y. Wu, Recent advances and future perspective on lignocellulose-based materials as adsorbents in diverse water treatment applications, Int. J. Biol. Macromol. 253 (2023) 126984.
[6] D. Zhang, J. Zuo, H. Xu, S. Gao, Y.-C. He, Pretreatment of wheat straw with alkali-assisted betaine-based deep eutectic solvents improves saccharification efficiency, Sustain. Chem. Pharm. 46 (2025) 102068.
[7] Y. Luo, Z. Li, X. Li, X. Liu, J. Fan, J.H. Clark, C. Hu, The production of furfural directly from hemicellulose in lignocellulosic biomass: a review, Catal. Today 319 (2019) 14–24.

[8] A. Kirui, W. Zhao, F. Deligey, H. Yang, X. Kang, F. Mentink-Vigier, T. Wang, Carbohydrate-aromatic interface and molecular architecture of lignocellulose, Nat. Commun. 13 (1) (2022) 538.
[9] R. Li, Y. Zheng, X. Zhao, Q. Yong, X. Meng, A. Ragauskas, C. Huang, Recent advances in biomass pretreatment using biphasic solvent systems, Green Chem. 25 (7) (2023) 2505–2523.
[10] P. Kalhor, K. Ghandi, Deep eutectic solvents for pretreatment, extraction, and catalysis of biomass and food waste, Molecules 24 (22) (2019) 4012.
[11] Q. Hu, Y. Xu, Y. Wang, W. Gong, C.-Y. Ma, S. Li, J.-L. Wen, Promoting the disassemble and enzymatic saccharification of bamboo shoot shells via efficient hydrated alkaline deep eutectic solvent pretreatment, Int. J. Biol. Macromol. 264 (2024) 130702.
[12] C. Cai, Y. Wang, W. Yu, C. Wang, F. Li, Z. Tan, Temperature-responsive deep eutectic solvents as green and recyclable media for the efficient extraction of polysaccharides from Ganoderma lucidum, J. Clean. Prod. 274 (2020) 123047.
[13] X. Li, W. Tang, Y.-C. He, Integrated understanding of acidic deep eutectic solvent choline chloride: oxalic acid pretreatment to enhance the enzymatic hydrolysis of rape straw, Ind. Crop. Prod. 206 (2023) 117691.
[14] W. Tang, B. Fan, X. Wang, C. Huang, Z. Tang, Y. He, Facilitating enzymatic hydrolysis efficiency of rape straw by pretreatment with a novel glycerol-based deep eutectic solvent, Ind. Crop. Prod. 206 (2023) 117587.
[15] G. Wu, Y. Cheng, C. Huang, C. Yong, Y. Fu, Deep eutectic solvent engineering: a novel ternary system for efficient lignocellulose extraction, Green Chem. 27 (5) (2025) 1556–1569.
[16] X. Zhao, G. Lyu, X. Meng, Y. Liu, Z. Wang, C.G. Yoo, Novel ternary deep eutectic solvent fractionation for effective utilization of willow, Bioresour. Technol. 407 (2024) 131148.
[17] Y. Xiong, W. Li, Z. Qin, T. Su, X. Xie, H. Ji, A green extraction technology of lignocellulose from cassava residue by mechanical activation-assisted ternary deep eutectic solvent, Int. J. Biol. Macromol. 281 (2024) 136339.
[18] Z. Jia, X. Li, S. Xu, Y. Sun, L. Chen, J. Ye, W. Li, T. Wu, J. Li, S. Wang, X. Xu, Novel ternary deep eutectic solvents pretreatment of corn stalk to realize high-value utilization, Chem. Eng. J. 500 (2024) 156567.
[19] R. Bai, W. Wang, M. Chen, Y. Wu, Study of ternary deep eutectic solvents to enhance the bending properties of ash wood, RSC Adv. 14 (12) (2024) 8090–8099.
[20] Y. Wang, J. Ryu, K.H. Kim, X. Meng, Y. Pu, Y. Tian, A. Eudes, G. Leem, A. J. Ragauskas, C.G. Yoo, Investigation of the effects of ternary deep eutectic solvent composition on pretreatment of sorghum Stover, AIChE J. 69 (12) (2023) e18227.
[21] Z. Wang, Y. Liu, K. Barta, P.J. Deuss, The effect of acidic ternary deep eutectic solvent treatment on native lignin, ACS Sustain. Chem. Eng. 10 (38) (2022) 12569–12579.
[22] R. Kandanelli, C. Thulluri, R. Mangala, P.V.C. Rao, S. Gandham, H.R. Velankar, A novel ternary combination of deep eutectic solvent-alcohol (DES-OL) system for synergistic and efficient delignification of biomass, Bioresour. Technol. 265 (2018) 573–576.
[23] X. Zhao, W. Tang, B. Fan, Y.-C. He, C. Ma, Implementing efficient and sustainable pretreatment of Sorghum stalks for delignification and xylan separation with a ternary deep eutectic solvent under mild conditions, Int. J. Biol. Macromol. 303 (2025) 140417.
[24] D. Zhang, J. Liu, H. Xu, H. Liu, Y.-C. He, Improving saccharification efficiency of corn Stover through ferric chloride-deep eutectic solvent pretreatment, Bioresour. Technol. 399 (2024) 130579.
[25] W. He, D. Yang, W. Tang, C. Ma, Y.-C. He, Integrated understanding of enhancing enzymatic hydrolysis of bulrush through the metal chloride-assisting choline chloride: glycerol pretreatment, Ind. Crop. Prod. 205 (2023) 117586.
[26] A.P. Abbott, R.C. Harris, K.S. Ryder, C. D'Agostino, L.F. Gladden, M.D. Mantle, Glycerol eutectics as sustainable solvent systems, Green Chem. 13 (1) (2011) 82–90.
[27] M. Park, J. Hwang, S. Jin, D. Jang, H.J. Kim, S.M. Choi, M.H. Seo, W.B. Kim, Unveiling the role of hydroxyl groups in glycerol as a critical descriptor for efficient electrocatalytic reforming of biomass molecules using PtCu alloy nanoparticle catalysts, Chem. Eng. J. 466 (2023) 143138.
[28] W. Wang, B. Zhu, Y. Xu, B. Li, H. Xu, Mechanism study of ternary deep eutectic solvents with protonic acid for lignin fractionation, Bioresour. Technol. 363 (2022) 127887.
[29] A.R. Mankar, A. Pandey, A. Modak, K.K. Pant, Pretreatment of lignocellulosic biomass: a review on recent advances, Bioresour. Technol. 334 (2021) 125235.
[30] P.J. Piedade, V. Venkat, K.W.A. Al-Shwafy, M.A. Aregawi, G. Dudek, M. Zygadło, R.M. Lukasik, Comprehensive wheat straw processing with deep eutectic solvent to deliver reducing sugar, Bioenergy Res. 17 (3) (2024) 1559–1568.
[31] F.H.B. Sosa, D.O. Abranches, A.M. da Costa Lopes, M.C. da Costa, J.A.P. Coutinho, Role of deep eutectic solvent precursors as Hydrotropes: unveiling synergism/antagonism for enhanced Kraft lignin dissolution, ACS Sustain. Chem. Eng. 12 (23) (2024) 8930–8940.
[32] Y. Zhang, W. Xie, W. Li, Y. Tang, J. Wu, L. Gan, L.-P. Xiao, X. Zhang, L. Lin, X. Tang, Acid-assisted pretreatment for sweet elephant grass in choline chloride/glycerol DES: experimental and mechanism study, Chem. Eng. J. 500 (2024) 157056.
[33] O. Yildirim, D. Tunay, B. Ozkaya, A. Demir, Optimization of oxalic and sulphuric acid pretreatment conditions to produce bio-hydrogen from olive tree biomass, Int. J. Hydrog. Energy 47 (62) (2022) 26316–26325.
[34] J. Mao, M. Zu, R. Wu, C. Xiao, J. Kang, Y. Fan, Study on the mechanism of lignin and lignin-carbohydrate complex hydrolysis under alkaline conditions by density functional theory, Biomass Bioenergy 185 (2024) 107233.
[35] T.R. Kozmelj, E. Bartolomei, A. Dufour, S. Leclerc, P. Arnoux, B. Likozar, E. Jasiukaitytė-Grojzdek, M. Grilc, Y. Le Brech, Oligomeric fragments distribution, structure and functionalities upon ruthenium-catalyzed technical lignin depolymerization, Biomass Bioenergy 181 (2024) 107056.
[36] X. Tan, C. Yang, Y. Li, Q. Zhang, Y. Sheng, C. Lai, Lignin fractionation and condensation in aromatic-additive-assisted acidic pretreatment and their influence on lignin's effect on the enzymatic hydrolysis, Int. J. Biol. Macromol. 295 (2025) 139471.
[37] H. Ji, Y. Song, X. Zhang, T. Tan, Using a combined hydrolysis factor to balance enzymatic saccharification and the structural characteristics of lignin during pretreatment of hybrid poplar with a fully recyclable solid acid, Bioresour. Technol. 238 (2017) 575–581.
[38] Y. Yuan, B. Jiang, H. Chen, W. Wu, S. Wu, Y. Jin, H. Xiao, Recent advances in understanding the effects of lignin structural characteristics on enzymatic hydrolysis, Biotechnol. Biofuels 14 (1) (2021) 205.
[39] R. Wang, L. Chen, J.Y. Zhu, R. Yang, Tailored and integrated production of Carboxylated cellulose nanocrystals (CNC) with Nanofibrils (CNF) through maleic acid hydrolysis, ChemNanoMat 3 (5) (2017) 328–335.
[40] W. Wang, Y. Xu, B. Zhu, H. Ge, S. Wang, B. Li, H. Xu, Exploration of the interaction mechanism of lignocellulosic hybrid systems based on deep eutectic solvents, Bioresour. Technol. 385 (2023) 129401.
[41] H.G.H. Yavuz, I. Yavuz, A. Isci, I. Turhan, Harnessing deep eutectic solvent for enhanced inulinase production from agricultural via submerged fermentation with aspergillus Niger, Int. J. Biol. Macromol. 295 (2025) 139592.
[42] X. Tian, X. Qin, X. Jia, Q. Lyu, S. Li, L. Jiang, L. Chen, Z. Yan, J. Huang, Lignocellulose degradation and temperature adaptation mechanisms during composting of mushroom residue and wood chips at low temperature with inoculation of psychrotolerant microbial agent, Environ. Pollut. 363 (2024) 125156.
[43] R.S. Abolore, S. Jaiswal, A.K. Jaiswal, Green and sustainable pretreatment methods for cellulose extraction from lignocellulosic biomass and its applications: a review, Carbohydrate Polymer Technologies and Applications 7 (2024) 100396.
[44] P. Ackerer, A. Bouissonnié, R. di Chiara Roupert, D. Daval, Estimating the activation energy of bond hydrolysis by time-resolved weighing of dissolving crystals, NPJ Mater. Degrad. 5 (1) (2021) 48.
[45] K. Wang, H. Guo, S. You, H. Yang, N. Li, X. Liang, Y. Qing, Y. Huang, Interpretive analysis of the relationship between pseudolignin structure and enzymatic hydrolysis, Ind. Crop. Prod. 214 (2024) 118519.
[46] V. Vorobyova, M. Skiba, G. Vasyliev, Deep eutectic solvents: quantum chemical investigation, thermal stability and physicochemical properties, Chem. Phys. 586 (2024) 112401.
[47] S. Sun, E. Li, X. Lin, X. Qiu, Elevating monosaccharides and monophenols production from wheat straw components via in-situ tailored lignin with deep eutectic solvents, Chem. Eng. Sci. 302 (2025) 120880.
[48] Y. Ma, Z. Liu, L. Lei, S. Hu, Y. Hou, Mechanism insights for the roles of hydrogen bonds on the dissolution and separation of lignin in LA-ChCl system, Ind. Crop. Prod. 222 (2024) 119911.
[49] M. Elsayed, M. Eraky, A.I. Osman, J. Wang, M. Farghali, A.K. Rashwan, I. H. Yacoub, D. Hanelt, A. Abomohra, Sustainable valorization of waste glycerol into bioethanol and biodiesel through biocircular approaches: a review, Environ. Chem. Lett. 22 (2) (2024) 609–634.
[50] H. Mahmood, S. Mehmood, A. Shakeel, T. Iqbal, M.A. Kazmi, A.R. Khurram, M. Moniruzzaman, Glycerol assisted pretreatment of lignocellulose wheat straw materials as a promising approach for fabrication of sustainable fibrous filler for biocomposites, Polymers 13 (2021) 388.
[51] K. Zheng, S. Hu, Z. Gong, M. Jia, K. Xu, J. Xu, L. Jiang, Y. Wang, S. Su, J. Xiang, Interaction among cellulose, hemicellulose and lignin during pressurized pyrolysis: importance of deoxygenation and aromatization reactions, Energy 314 (2025) 134320.
[52] T. Jeoh, C.I. Ishizawa, M.F. Davis, M.E. Himmel, W.S. Adney, D.K. Johnson, Cellulase digestibility of pretreated biomass is limited by cellulose accessibility, Biotechnol. Bioeng. 98 (1) (2007) 112–122.
[53] C. Cai, C. Zhang, N. Li, H. Liu, J. Xie, H. Lou, X. Pan, J.Y. Zhu, F. Wang, Changing the role of lignin in enzymatic hydrolysis for a sustainable and efficient sugar platform, Renew. Sust. Energ. Rev. 183 (2023) 113445.
[54] L. Wang, G. Li, X. Chen, Y. Yang, R.K. Liew, H.M. Abo-Dief, S.S. Lam, R. Sellami, W. Peng, H. Li, Extraction strategies for lignin, cellulose, and hemicellulose to obtain valuable products from biomass, Adv. Compos. Hybrid Mater. 7 (6) (2024) 219.
[55] H. Ma, P. Fu, J. Zhao, X. Lin, W. Wu, Z. Yu, C. Xia, Q. Wang, M. Gao, J. Zhou, Pretreatment of wheat straw lignocelluloses by deep eutectic solvent for lignin extraction, Molecules 27 (22) (2022) 7955.
[56] J. Xie, S. Liu, Kinetic understanding of fiber surface lignin effects on cellulase adsorption and hydrolysis, Results Surf Interfaces 14 (2024) 100185.
[57] K.S. Salem, N.K. Kasera, M.A. Rahman, H. Jameel, Y. Habibi, S.J. Eichhorn, A. D. French, L. Pal, L.A. Lucia, Comparison and assessment of methods for cellulose crystallinity determination, Chem. Soc. Rev. 52 (18) (2023) 6417–6446.
[58] K. Li, H. Yang, L. Jiang, X. Liu, P. Lang, B. Deng, N. Li, W. Xu, Glycerin/NaOH aqueous solution as a green solvent system for dissolution of cellulose, Polymers 12 (8) (2020) 1735.
[59] H. Guo, Y. Zhao, J.-S. Chang, D.-J. Lee, Enzymes and enzymatic mechanisms in enzymatic degradation of lignocellulosic biomass: a mini-review, Bioresour. Technol. 367 (2023) 128252.
[60] J. Cheng, X. Liu, C. Huang, Y. Zhan, C. Huang, T. Chen, X. Meng, C.G. Yoo, G. Fang, A.J. Ragauskas, Novel biphasic DES/GVL solvent for effective biomass fractionation and valorization, Green Chem. 25 (16) (2023) 6270–6281.

[61] Y. He, R. Shen, L. Shao, C. Wang, G. Yang, F. Xu, Efficient binary diol-based deep eutectic solvent pretreatment for enhancing enzymatic saccharification and lignin fractionation from eucalyptus, Bioresour. Technol. 412 (2024) 131423.
[62] Q. Yang, W. Tang, L. Li, M. Huang, C. Ma, Y.-C. He, Enhancing enzymatic hydrolysis of waste sunflower straw by clean hydrothermal pretreatment, Bioresour. Technol. 383 (2023) 129236.
[63] L. Zhu, W. Tang, C. Ma, Y.-C. He, Efficient co-production of reducing sugars and xylooligosaccharides via clean hydrothermal pretreatment of rape straw, Bioresour. Technol. 388 (2023) 129727.
[64] F. Li, Q. Li, J. Lv, M. Huang, Z. Ling, Y. Meng, F. Chen, Z. Ji, A novel seawater hydrothermal-deep eutectic solvent pretreatment enhances the production of fermentable sugars and tailored lignin nanospheres from Pinus massoniana, Int. J. Biol. Macromol. 267 (2024) 131596.
[65] X. Yin, T. Cai, C. Liu, Y. Ma, J. Hu, J. Jiang, K. Wang, A novel solvothermal biorefinery for production of lignocellulosic xylooligosaccharides, fermentable sugars and lignin nano-particles in biphasic system, Carbohydr. Polym. 295 (2022) 119901.
[66] X. Wei, Y. Wang, J. Chen, F. Xu, Z. Liu, X. He, H. Li, Y. Zhou, Adsorption of pharmaceuticals and personal care products by deep eutectic solvents-regulated magnetic metal-organic framework adsorbents: performance and mechanism, Chem. Eng. J. 392 (2020) 124808.
[67] M. Khnifira, W. Boumya, J. Atarki, M. Sadiq, M. Achak, A. Bouich, N. Barka, M. Abdennouri, Experimental, DFT and MD simulation combined studies for the competitive adsorption of anionic and cationic dyes on activated carbon in an aqueous medium, J. Mol. Struct. 1310 (2024) 138247.
[68] C. Lefebvre, G. Rubez, H. Khartabil, J.-C. Boisson, J. Contreras-García, E. Hénon, Accurately extracting the signature of intermolecular interactions present in the NCI plot of the reduced density gradient versus electron density, Phys. Chem. Chem. Phys. 19 (27) (2017) 17928–17936.
[69] T. Lu, Q. Chen, Independent gradient model based on Hirshfeld partition: a new method for visual study of interactions in chemical systems, J. Comput. Chem. 43 (8) (2022) 539–555.
[70] X. Li, Y. Wang, W. Tang, B. Fan, Y.-C. He, C. Ma, Co-production of reducing sugars and xylo-oligosaccharides from rape straw through sodium dodecyl sulfate-mediated p-toluenesulfonic acid pretreatment, Ind. Crop. Prod. 222 (2024) 119888.